1. Clinical Question & Core Hypothesis

Clinical Question

Ground-glass–like imaging features or stromal-rich histology are imperfect predictors of clinical outcome — why does an imaging/stromal signature of invasiveness sometimes fail to portend poor survival? 2) Tier 1 paths used: EMT (loss of epithelial identity), E-cadherin (adhesion loss), and TP53 (tumor-suppressor dysfunction) form a mechanistic backbone linking epithelial plasticity to stromal remodeling and invasive behavior; I built on these because EMT-driven E-cadherin downregulation plausibly increases stromal engagement (raising TSR) while TP53 mutation can shift the biology from a reactive/desmoplastic state to aggressive invasion. 3) Tier 2 evidence (CIS, Klebsiella, invasive cancer) supports the directionality that preinvasive states, microbial/ inflammatory triggers, and true invasive conversion can modulate the TSR–outcome relationship, motivating inclusion as secondary modifiers. 4) Endpoint choice: Overall survival (OS) was chosen because it is the most definitive clinical outcome available in TCGA-BRCA and best captures the net impact of invasion, host response, and molecular drivers. 5) Deliberately excluded: Ground glass nodules (GGNs) were not chosen as the imaging feature because the permitted imaging features list for this dataset does not include GGNs; instead I used stromal-to-tumor ratio (TSR) as the histologic/imaging-relevant proxy for stromal/reactive phenotypes. I excluded broad cross-tissue elements (VVF, bladder-specific cohort nuances) from the core hypothesis to keep the mechanistic question sharp and testable in breast TCGA data?

Working Hypothesis

In breast cancer, the prognostic impact of a high stromal-to-tumor ratio (TSR) on overall survival (OS) is conditional on epithelial–mesenchymal transition and genomic context: TSR-high tumors with loss of E-cadherin function (low CDH1 expression or E-cadherin dysfunction) and TP53 mutation will exhibit the worst OS, whereas TSR-high tumors with intact TP53 and preserved E-cadherin will show attenuated or no excess mortality risk. Formally, there will be a significant three-way interaction on OS: TSR_high × TP53_mutant × E-cadherin_low predicting higher hazard than TSR_high × TP53_wild-type × E-cadherin_preserved, with intermediate risk in other combinations. Secondary endpoints include progression-free interval (PFI) to assess time-to-event dynamics. The hypothesis will be tested first in a large breast cancer cohort with available TSR measurements (imaging/histology proxy), TP53 mutation status, and E-cadherin/CDH1 status, and then validated in an independent cohort. Analyses will adjust for key covariates (age, stage, grade, molecular subtype) and will explore potential effect modification by molecular subtype. Secondary modifiers—Klebsiella, CIS, and invasive cancer—will be examined as contextual factors that may modulate the TSR–OS relationship. Mechanistic follow-up studies (in vitro/in vivo) will probe whether TP53 mutation and E-cadherin loss synergistically enhance EMT/invasion in TSR-high contexts, supporting causality beyond association.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, E-cadherin
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 (tumor-suppressor dysfunction) form a mechanistic backbone linking epithelial plasticity to stromal remodeling and invasive behavior; I built on these because EMT-driven E-cadherin downregulation plausibly increases stromal engagement (raising TSR) while TP53 mutation can shift the biology from a reactive/desmoplastic state to aggressive invasion.

E-cadherin: E-cadherin (adhesion loss), and TP53 (tumor-suppressor dysfunction) form a mechanistic backbone linking epithelial plasticity to stromal remodeling and invasive behavior; I built on these because EMT-driven E-cadherin downregulation plausibly increases stromal engagement (raising TSR) while TP53 mutation can shift the biology from a reactive/desmoplastic state to aggressive invasion.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Klebsiella	Supporting modifier from knowledge graph
Secondary	CIS	Supporting modifier from knowledge graph
Secondary	Invasive cancer	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Klebsiella: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CIS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.